

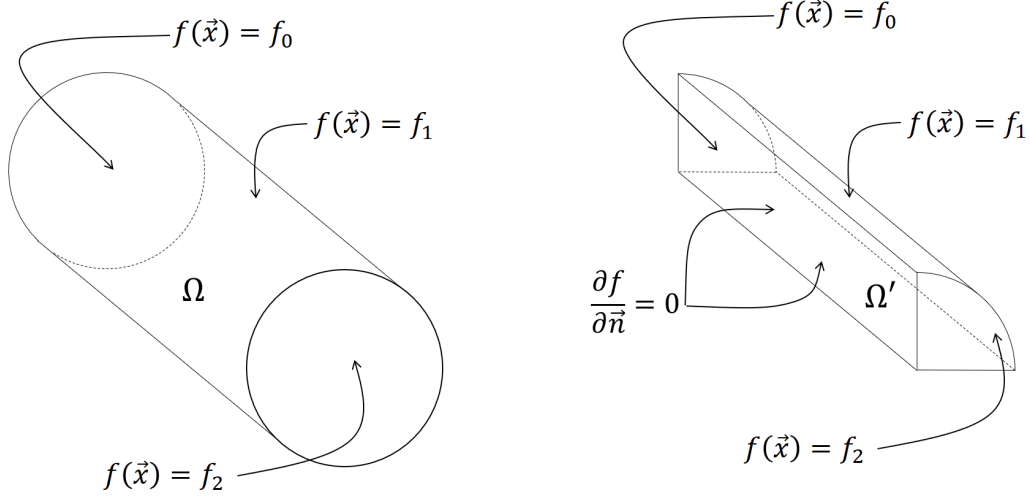


Figure 1: Example of a domain reduction using known internal symmetries.

on the border $\partial\Omega$ [1], having for instance the form

$$a_{ij} \frac{\partial^2 f}{\partial x_i \partial x_j} = g, \quad (1)$$

where a_{ij} and g are known functions of $\vec{r}(x_1, x_2, x_3)$ and $f(\vec{r})$ is the function searched on Ω . Usual solving methods are the finite element (FEM) [2] and the finite difference (FDM) [3]. They both rely on the discretization of space forming a mesh of N points $M_n, n = 1, \dots, N$ covering Ω and the linearization of the PDE to solve. The methods lead to the build up of a linear problem

$$AX = b, \quad (2)$$

where X is the unknown vector of component $f(M_n), n = 1, \dots, N$, b is a constant vector defined by specific values given on Ω and $\partial\Omega$ and A is a square matrix of rank N . The boundary conditions (BC) on $\partial\Omega$ are usually of Dirichlet type ($\mathcal{D}$), when f is known, or Neumann ($\mathcal{N}$), when $\partial f / \partial \vec{n}$ is known ($\vec{n}$ normal to $\partial\Omega$), or a combination of the two. When Ω is large or the need for accuracy is high, N becomes rapidly very large ($N \gg 10^6$) and the computer solving and the numerical storing of the inverse matrix A^{-1} , solution of Eq. 2, can become problematic. It is generally of interest to reduce the problem size by considering all the symmetries applying both on Ω , $\partial\Omega$ and g . An example of such a domain reduction from Ω down to